Intelligence in Crisis: Multi-Agent AI Systems for Emergency Response Decision-Making Using Large Language Models and Parallel Agents

Project Title and Faculty Mentor

Intelligence in Crisis: Multi-Agent AI Systems for Emergency Response Decision-Making Using Large Language Models and Parallel Agents

Faculty Mentor:
Dr. Mubarak Shah
Director, Institute for AI
Trustee Chair Professor
Department of Computer Science
College of Engineering & Computer Science
University of Central Florida

Project Objective or Aim

This project aims to explore how multi-agent artificial intelligence systems can be used to simulate crisis scenarios and improve emergency response decision-making. Specifically, the study will investigate how multiple AI agents (an individual AI entity assigned to a specific task that can interact with other agents), each representing different roles such as civilians, responders, and government officials, interact within a dynamic disaster environment. The central research question is: how can multi-agent AI simulations enhance the speed, coordination, and effectiveness of decision-making during high-pressure crisis situations? By developing and evaluating a prototype simulation system powered by large language models and parallel agent execution, this project seeks to better understand how AI-driven simulations can support training, planning, and real-time strategy development for emergency management.

Project Background and Significance

Emergency response scenarios, such as natural disasters, public health crises, and large-scale accidents, require rapid and coordinated decision-making under conditions of uncertainty. In these situations, delays or poor coordination can lead to significant loss of life, inefficient allocation of resources, and long-term societal consequences. Traditional training approaches, including tabletop exercises and static simulations, often fail to capture the complexity and unpredictability of real-world crises, limiting their effectiveness. As a result, there is a need for more adaptive and realistic systems that can better model dynamic environments and support decision-making under pressure.

Recent advancements in artificial intelligence, particularly in large language models and multi-agent systems, offer promising solutions to this problem. A multi-agent system refers to a collection of independent entities that interact within a shared environment to achieve individual or collective goals (Wooldridge, 2009). These systems are capable of modeling complex interactions and emergent behaviors, making them well-suited for simulating dynamic environments (Jennings, 2000). Additionally, large language models enable agents to generate context-aware responses and simulate realistic communication, allowing for more human-like decision-making processes (Russell & Norvig, 2021).

This project is grounded in the theoretical framework of multi-agent systems and computational simulation. Multi-agent system theory focuses on how independent agents interact to produce coordinated or emergent outcomes within complex environments. In addition, this research draws on principles from human behavior modeling and decision-making frameworks commonly used in emergency response simulations (Pan, 2016). These frameworks guide both the design of the simulation system and the evaluation of how agents communicate, coordinate, and respond to changing conditions.

Despite these developments, there remains a gap in applying multi-agent AI systems to emergency response training and decision support. Existing simulation tools often lack the ability to model realistic communication between entities and adapt to rapidly changing conditions. This project seeks to address this gap by developing a simulation framework that integrates autonomous decision-making with natural language interaction, enabling more flexible and realistic representations of crisis scenarios.

The significance of this research lies in its potential to improve both preparedness and real-time response. By creating more interactive and realistic simulations, this project can provide valuable insights into decision-making strategies, coordination patterns, and system-level behavior during crises. Furthermore, it contributes to the broader field of artificial intelligence by demonstrating how multi-agent systems and language models can be applied to real-world challenges. This research also has the potential to benefit the University of Central Florida community by fostering interdisciplinary collaboration and enhancing educational tools related to public safety, engineering, and computer science.

Research Methods

This project will be carried out through the design, development, and evaluation of a multi-agent AI simulation system designed for emergency response scenarios. The methodology is structured into three main phases: system development, simulation testing, and performance evaluation. Each phase builds on the previous one to ensure a clear and organized approach to developing and assessing the system.

In the first phase, a prototype simulation platform will be developed to model a dynamic crisis environment using multiple autonomous AI agents. Each agent will represent a specific role, such as emergency responders, civilians, or coordinating agencies, and will be able to communicate and make decisions using natural language powered by large language models. The system will use a lightweight full-stack architecture, including a backend service to manage agent coordination and a frontend interface to visualize interactions and system behavior. This phase is essential for establishing the core functionality and structure of the simulation.

In the second phase, the system will be tested using predefined crisis scenarios that vary in complexity and scale. These scenarios will allow for observation of how agents behave under different levels of pressure and uncertainty. Key variables, including response time, communication efficiency, and coordination effectiveness, will be recorded during each simulation. Multiple simulation runs will be conducted to ensure consistency and reliability of results. The collected data will then be quantitatively analyzed to compare performance across different conditions and identify trends in agent behavior.

In the final phase, results will be analyzed to evaluate overall system effectiveness and identify patterns in decision-making and coordination. Based on these findings, the system may be refined to improve performance and interaction between agents. The project will follow a structured semester timeline, with weeks 1–3 focused on research and system design, weeks 4–8 on development, weeks 9–12 on testing and data collection, and weeks 13–15 on analysis and final reporting. This timeline ensures that the project remains feasible and well-organized within a single academic term.

Expected Outcome

The primary outcome of this project will be the development of a functional multi-agent AI simulation platform designed to model emergency response scenarios. This system will serve as an interactive prototype that demonstrates how artificial intelligence can simulate complex decision-making processes in dynamic and uncertain environments. Users will be able to observe how different agents, such as emergency responders, civilians, and coordinating agencies, interact, communicate, and respond to evolving conditions. By modeling these interactions, the system will provide a clearer understanding of how different strategies influence outcomes during a crisis.

In addition to the working prototype, the project will produce a detailed report that documents the system design, development process, and evaluation results. This report will focus on key findings related to coordination efficiency, communication patterns, and overall decision-making effectiveness observed during simulations. The project will also include a visual interface that allows users to explore simulation results, making it easier to analyze system behavior and compare different scenarios. These outputs will help demonstrate both the technical capabilities of the system and its practical value.

The results of this project will be shared with others through a university research event at the University of Central Florida, where the system may be presented in the form of a research poster or live demonstration. This will allow the project to reach a broader academic audience and encourage feedback from students and faculty across multiple disciplines.

This research contributes to the field by exploring practical applications of multi-agent AI in crisis response and decision support. It provides a foundation for future development of more advanced simulation tools that can be used for training and planning. In the long term, this work has the potential to improve preparedness and response strategies by offering more realistic and interactive ways to model real-world emergency situations.

Literature Review

Wooldridge, M. (2009). *An introduction to multiagent systems* (2nd ed.). Wiley.

Russell, S., & Norvig, P. (2021). *Artificial intelligence: A modern approach* (4th ed.). Pearson.

Jennings, N. R. (2000). On agent-based software engineering. *Artificial Intelligence*, 117(2), 277–296.

Pan, X. (2016). Computational modeling of human behavior in emergency evacuation: A review. *Physica A: Statistical Mechanics and Its Applications*, 463, 299–312.

Silverman, B. G., Hanrahan, N., Bharathy, G., Gordon, K., & Johnson, D. (2006). A systems approach to healthcare: Agent-based modeling, simulation, and decision support. *Proceedings of the Winter Simulation Conference*, 122–131.

Preliminary Work and Experience

I have experience developing AI-driven systems and full-stack applications that closely align with the goals of this project. Most notably, I contributed to the development of a multi-agent crisis simulation platform as part of UCF's premier hackathon, KnightHacks. In this project, I focused on backend development and agent coordination, helping design a system where multiple AI agents interact within a dynamic environment. The project earned Best Overall Hack and placed second in the Google ADK Agentic Challenge, highlighting both its technical strength and practical relevance.

This hackathon project, titled "Emergent," serves as an early prototype for the proposed research. It demonstrates the feasibility of using multi-agent AI systems to simulate complex crisis scenarios and supports the potential for further development into a more robust platform. In addition to its success at KnightHacks, the project was also recognized as a winner in the UCF Innovation Challenge, which further validated its value as a product and its potential impact for real-world stakeholders.

I have also worked as a Software Engineer Intern at Siemens Energy, where I developed internal tools used to support engineering workflows and system analysis. This experience strengthened my ability to design scalable systems.

IRB/IACUC Statement

This project does not involve human or animal subjects; therefore, IRB and IACUC approval are not required.

Budget

Item	**Cost**	**Justification**
Cloud Computing	$600	GPU resources for running large language models and multi-agent simulations.
Large Language Model API Usage	$250	API costs for agent communication and decision-making.
Backend Hosting and Database (Supabase or cloud storage)	$200	Storage for simulation data, logs, and results.
Development and Deployment Tools	$150	Domain registration, hosting services, and deployment tools.
Testing and System Optimization	$150	Supports running multiple simulations and improving system performance.
Dissemination (Poster printing and materials)	$100	Materials to present the project at a university research event.
Total	**$1,350**	